Desperate for shade on your walk? There's (almost) an app for that.

By mapping shade, a new online tool calculates the best way to stroll a city without overheating.

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It's getting increasingly unbearable, even downright dangerous, to walk in cities. That's because of the urban heat island effect: Buildings, sidewalks, and roads absorb the sun's energy and radiate it back at pedestrians, raising temperatures far above that you'd find in the surrounding countryside. If a city like Phoenix doesn't have enough shade, people can't move safely by foot, when really we need to help folks do more of that, because ambling improves public health and reduces vehicle traffic.

If you boot up a maps app on your phone, it provides the most efficient way of getting from point A to point B in a metropolis, but it tells you nothing about the blast furnace you'll endure along the way. A new research project from Arizona State University hopes to fix that with Cool Routes, an online tool that calculates the heat that you might feel along active mobility paths. In addition to finding the shortest path, it determines the coolest and shadiest — and therefore safest, thermally speaking — one. Though it's limited at the moment to the ASU Tempe campus, the researchers are open-sourcing the tool for any city to use, and maybe one day popular map apps will incorporate such data too.

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Your weather app isn't lying about the warmth, but it doesn't provide a full picture about the heat load on the human body. Air temperature is just one component of how comfortable you feel: Add high humidity, and 80 degrees Fahrenheit feels more like 100, because the extra atmospheric moisture makes sweating less efficient at cooling the body. A lack of shade makes the heat feel even worse. “That is what makes being in a hot environment dangerous, because we see lower numbers on our phones,” said Isaac Buo, an urban informatics scientist at ASU who co-led Cool Routes with Ariane Middel, director of the university’s SHaDE Lab. Take a diversion into the shade and you reduce the heat load by half.

Instead of simply referencing a thermometer, Cool Routes calculates “mean radiant temperature.” It, for instance, determines the shade available from buildings and trees. (That’s possible thanks to the United States Geological Survey, which has used lidar technology to map landscapes in extreme detail.) “It takes a high-resolution representation of the urban environment, and also the forecasted weather data, for us to simulate the thermal conditions at any given time,” Buo said.

That time is an important consideration, because walking to work at 8 a.m. will feel quite different than walking to lunch at noon, and not just because temperatures rise throughout the day. At high noon, for example, skyscrapers might not provide much shade because the sun is directly overhead, but trees will continue to because they create an overhanging canopy. This complexity of the urban environment is why heat exposure differs dramatically not just neighborhood to neighborhood, but even block to block.

To account for this, Cool Routes factors in heat coming at pedestrians from six directions at any given point and time: From north, south, east, and west, and from above and below. The ASU researchers ground-truthed these calculations with a “mobile human-biometeorological cart” called MaRTy (MRT coming from mean radiant temperature), which they rolled along courses suggested by Cool Routes on typical hot summer days.

In addition to individuals using Cool Routes to find the most comfortable course, cities might use the platform to determine where to prioritize tree planting and create parks. Maybe there’s a particularly popular route that people follow from a subway station to, say, a financial district, where adding vegetation might reduce temperatures by several degrees. (Really, trees are one of the simplest yet most powerful tools for any city: In addition to providing cooling, green patches provide habitat for animals and absorb stormwater to reduce the risk of flooding.) And even though at its core it’s meant to encourage walking, the platform could further support public transportation systems by identifying where bus stops could use some shade for people to wait under.

While it’s early days for Cool Routes, a future update for map apps might incorporate this kind of data. Like they already suggest walking routes with fewer hills, for example, they might one day suggest ones with less heat. “If you’re willing to make a detour of, say, two extra minutes,” Buo said, “we can get you through a route that is well-shaded.” How cool is that?

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